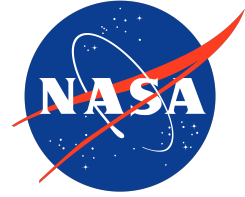




ASIO
Interface Control Document (ICD)



Date: 10/29/2025

Document Number: ASIO-SYSE-0001

MUSE - ASIO Interface Control Document



MUSE - ASIO Interface Control Document

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1.3	Updated software section, Updated electrical connector diagrams	10/29/2025	Aidan Rasmussen

TBX Log

Page	Comments	Actions
19	Final filter installation	LMSAL & MSU will decide upon final filter installtion timing after CDR
20	Post-shipment performance verification plan	MSU to develop comprehensive performacne test to be use for performance verification post CDR
51	MSU Interface Drawings Configuration and Joint Agreement	LMSAL & MSU to agree on drawing number

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LIST OF ACRONYMS

ASIO	Active Sun Irradiance Observer
CBE	Current Best Estimate
DA	Detector Assembly
EMI	Electromagnetic Interference
FEA	Finite Element Analysis
FEM	Finite Element Model
GSE	Ground Support Equipment
ICD	Interface Control Document
LMSAL	Lockheed Martin Solar and Astrophysics Laboratory
MSU	Montana State University
MUSE	Multi-slit Solar Explorer
NTE	Not to Exceed
SPA	Shadow Plate Assembly
SSEL	Space Science and Engineering Laboratory

1 Introduction

1.1 Scope

This document defines the interfaces of the Active Sun Irradiance Observer (ASIO), a student science collaboration that is part of the MUSE (MULTi-slit Solar Explorer) project.

This document contains information regarding mechanical/electrical hardware and software interfaces, thermal requirements, environments, and contamination control. Its primary function is to provide interface controls for each MSU-provided component and mating interface with MUSE.

This ICD is a living document that will be continuously updated as needed to document the Detector Assembly (DA) and Shadow Plate Assembly (SPA) interfaces between MSU and LMSAL.

1.2 Instrument Description

ASIO is a three-channel radiometer sensitive to extreme ultraviolet (EUV), soft X-ray (SXR), and hard X-ray (HXR) emissions. The instrument will measure disk-integrated solar flare emissions at a rapid cadence with SXR source locating capability to accurately track hot spots during the evolution of the flare. ASIO will investigate if there are sub-second, quasi-periodic pulsations (QPPs) during C5+ solar flares. ASIO is self-contained and external to MUSE instrument and spacecraft.

1.2.1 Planned Testing

The flight load envelope for structural design, vibration spectrum for vibration testing, and thermal envelope are described below. Referenced documents are pulled from the provided MUSE Observatory Environmental Requirements CDRL 27.

ASIO will be designed and tested for vibration testing per GEVS 2.4.2.5, as seen in Figure 1. Section 3.8.3 of MUSE-RQ-18-0039, the instrument shall be screened for design and workmanship defects, and components/units shall be subjected to a random vibration test along each of three mutually perpendicular axes. In addition, when components are particularly sensitive to the acoustic environment, an acoustic test shall be considered.

Generalized Random Vibration Test Levels
Components (ELV)
22.7-kg (50-lb) or less

Frequency (Hz)	ASD Level (g^2/Hz)	
	Qualification	Acceptance
20	0.026	0.013
20-50	+6 dB/oct	+6 dB/oct
50-800	0.16	0.08
800-2000	-6 dB/oct	-6 dB/oct
2000	0.026	0.013
Overall	14.1 G_{rms}	10.0 G_{rms}

The acceleration spectral density level may be reduced for components weighing more than 22.7-kg (50 lb) according to:

	Weight in kg	Weight in lb	
dB reduction	$= 10 \log(W/22.7)$	$10 \log(W/50)$	
ASD(50-800 Hz)	$= 0.16 \cdot (22.7/W)$	$0.16 \cdot (50/W)$	for protoflight
ASD(50-800 Hz)	$= 0.08 \cdot (22.7/W)$	$0.08 \cdot (50/W)$	for acceptance

Where W = component weight.

The slopes shall be maintained at + and - 6dB/oct for components weighing up to 59-kg (130-lb). Above that weight, the slopes shall be adjusted to maintain an ASD level of 0.01 g^2/Hz at 20 and 2000 Hz.

For components weighing over 182-kg (400-lb), the test specification will be maintained at the level for 182-kg (400 pounds).

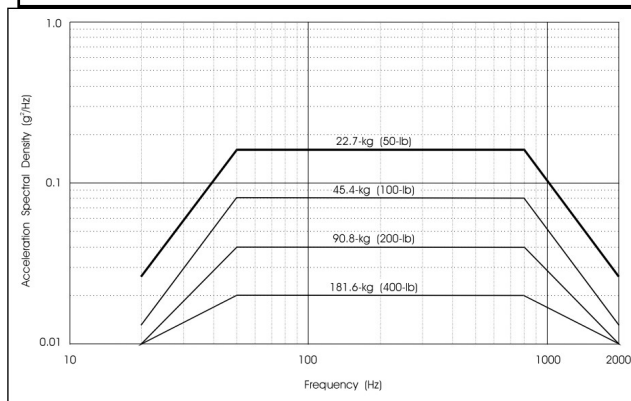


Figure 1: GEVS 2.4.2.5

For EMI testing the MUSEENV31242 indicates all subsystems and instruments electronics shall not exhibit any malfunction, degradation of performance, or deviation from specific indications, beyond the tolerances indicated equipment or subsystem specification when subjected to the CS116 test. For MUSE, the values of I_p and I_{max} shall be 1.0 Amps. Figure 2 details the CS116 test limit.

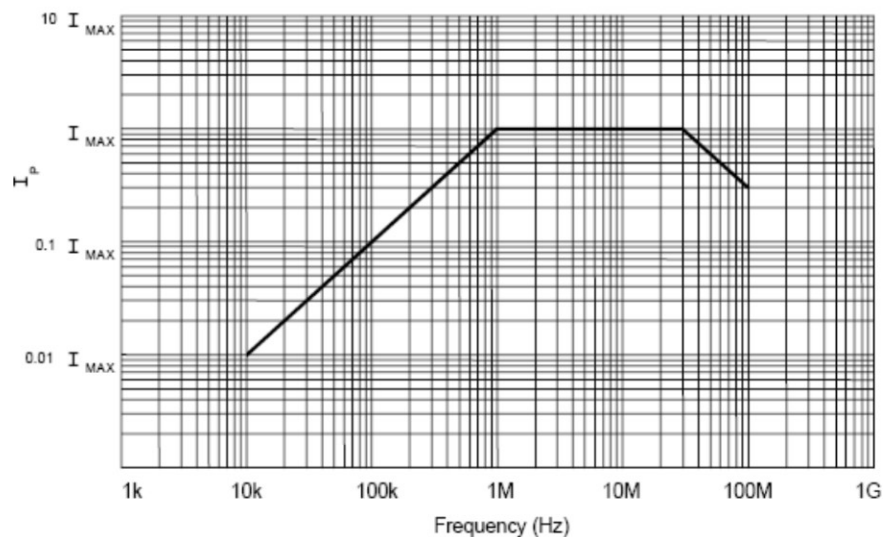


Figure 2: CS116

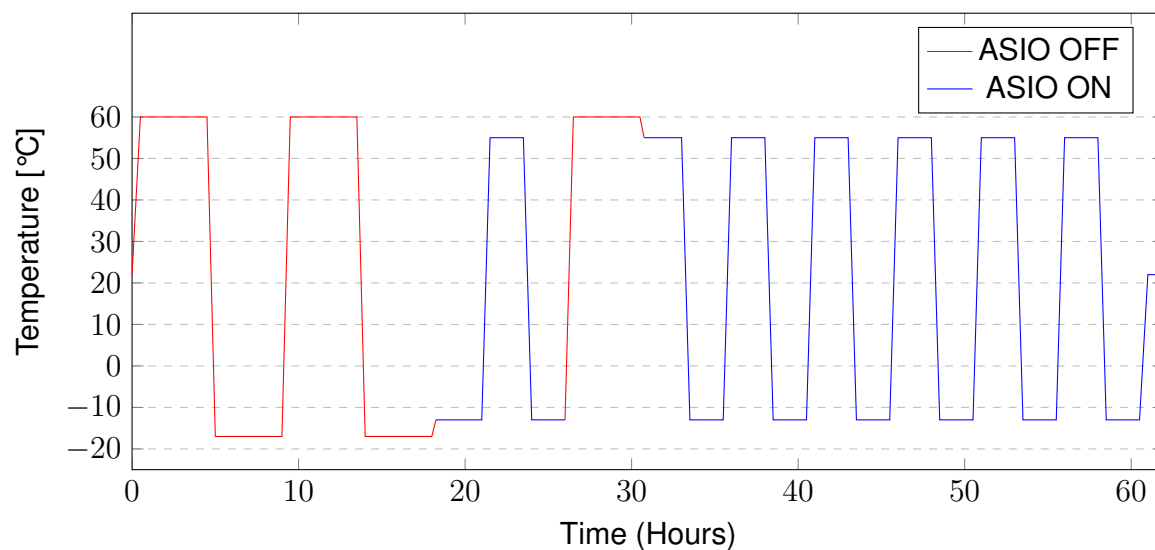
GSFC-STD-7000B and MSFC-SPEC-1238 outline recommendations for thermal testing. Thermal vacuum cycling on ASIO will be completed to demonstrate the performance of the flight hardware and provide Environmental Stress Screening, over a temperature range extended just beyond that which the flight hardware is allowed to operate over the mission timeline. ASIO only has two modes of operation, on and off. ASIO will be tested as a workmanship test that is meant to prove that ASIO will survive when exposed to operational and survival limits. As such, the thermal cycling profile in Table [1] includes hot start-up and cold start-up cycles.

ASIO will be tested under GSFC-STD-7000B recommendations, this includes two survival thermal cycles with 4-hour dwells at each extreme. Additionally, a total of six operational cycles with 2-hour dwells at each extreme. ASIO will be cycled during the transitions between thermal extremes while the internal temperature is within operational limits.

Thermal vacuum cycling will not occur until the vacuum chamber pressure of $1 * 10^{-5}$ torr or less is reached.

The operational test temperature range for ASIO is 55°C to -13°C and the survival temperature range is 60°C to -17°C.

Table 1: ASIO Thermal Cycling Temperature Profile



1.2.2 Instrument Coordinate System

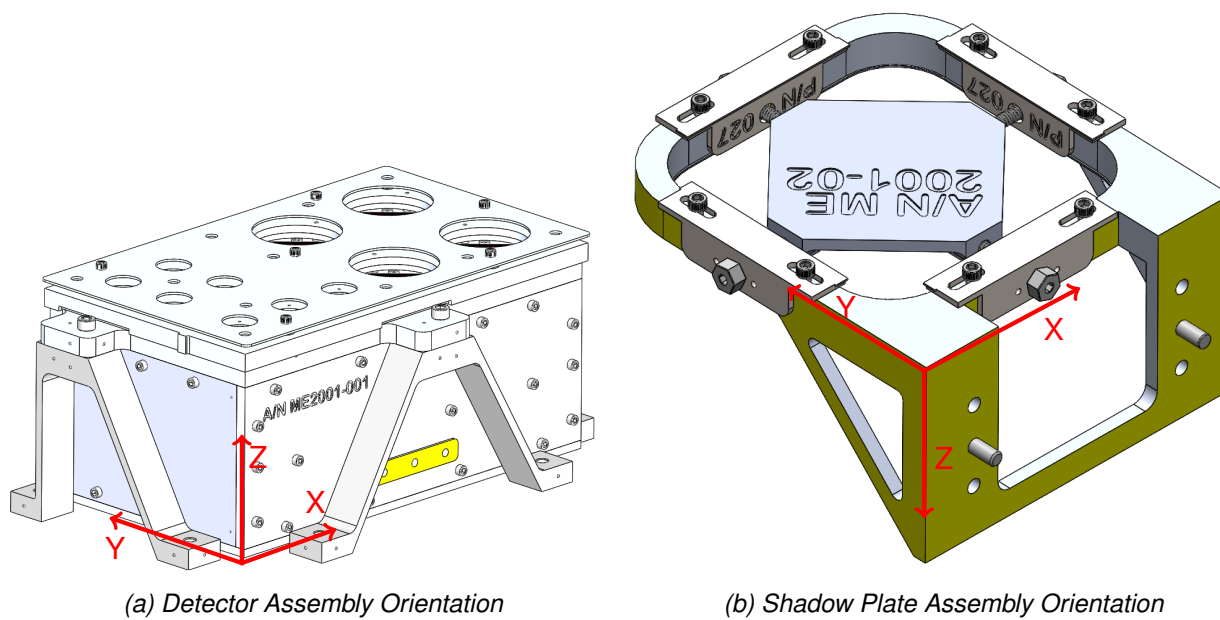


Figure 3: ASIO Orientation

1.2.3 Photodiode Coordinate System

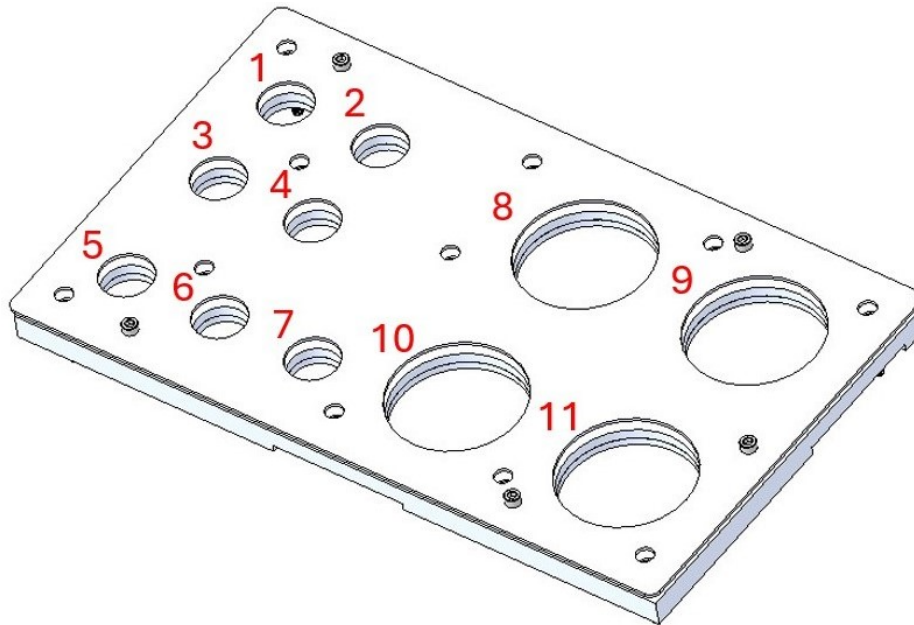


Figure 4: Photodiode Orientation

2 Applicable Documents

The following documents form a part of this ICD to the extent specified herein. In the event of a conflict between a referenced document and this ICD, the ICD takes precedence. References to the documents contained in the body of this ICD are by basic number only (i.e. without revision designation). This section controls the revision of referenced documents. A revision status of NR means the document was not released prior to this revision of the ICD. Information provided in a Mechanical Interface Drawing (MID) or Mechanical Interface Control Document supersedes specific piece-part or subassembly drawings.

All document names have been changed to ASIO. These documents are the same as the old versions with the name BUBO, as long as the revision number is the same. BUBO documents can be referenced unless there is a new revision number.

2.1 MSU Documents

Table 2: Applicable documents from MSU.

Document Number	Document Title	Applicability
ME1005-001	DETECTOR ASSEMBLY	Envelope, interface dimensions, thermal hardware, field of view and grounding interface for detector assembly
ME1005-002	SHADOW PLATE ASSEMBLY	Envelope and interface dimensions for shadow plate assembly
ASIO-SYSE-0002	MUSE - ASIO Mechanical Interface Control Document	Mechanical Interfaces, Fastener Responsibility, Surface Treatments, Mass Allocation
ASIO-SYSE-0014	ASIO Comprehensive Performance Test	Reference
ASIO-SYSE-0015	ASIO Structural Analysis Report	Static, Modal, Random Vibration, Venting, and Acoustic Analyses and Margins
MUSE0001449	ASIO Power System Diagram	Power system circuit diagram

2.2 LMSAL Documents

Table 3: Applicable documents from LMSAL.

Document Number	Document Title	Applicability
MUSE-RQ-18-0039	MUSE Observatory Environmental Requirements CDRL 27	Establishing ASIO Requirements
MUSE-PN-18-0078	MUSE Contamination Control Plan (CCP) CDRL35	Contamination Control Guidelines for ASIO
MUSE-PN-18-0066	MUSE Mission Assurance Implementation Plan	Contamination Control Requirements

2.3 Other Documents

Table 4: Other applicable documents.

Document Number	Document Title	Applicability
GSFC-STD7000B	General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects	Verification Standards for ASIO
MSFC-SPEC-1238	Thermal Vacuum Bakeout Specification For Contamination Sensitive Hardware	TVac Bakeout Standards for ASIO
MSFC-STD-486B	Standard, Threaded Fasteners, Torque Limits For	Standard torques specs for fasteners for ASIO
978-1-884989-11-7	Spacecraft Thermal Control Handbook	Optical properties of materials

3 MECHANICAL

3.1 Interfacing Parts

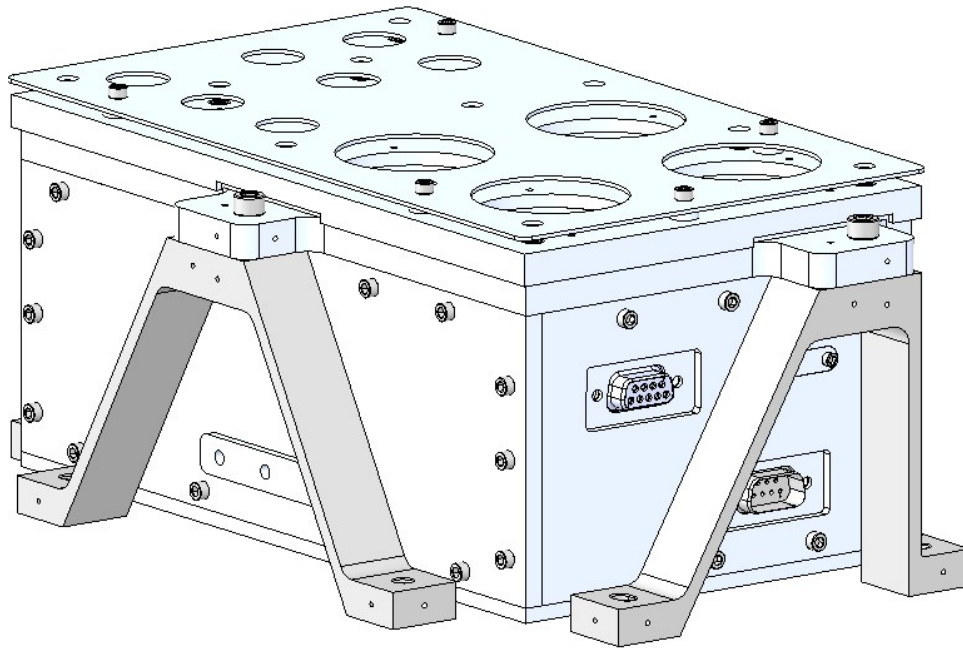


Figure 5: 3D Rendering of ASIO Instrument

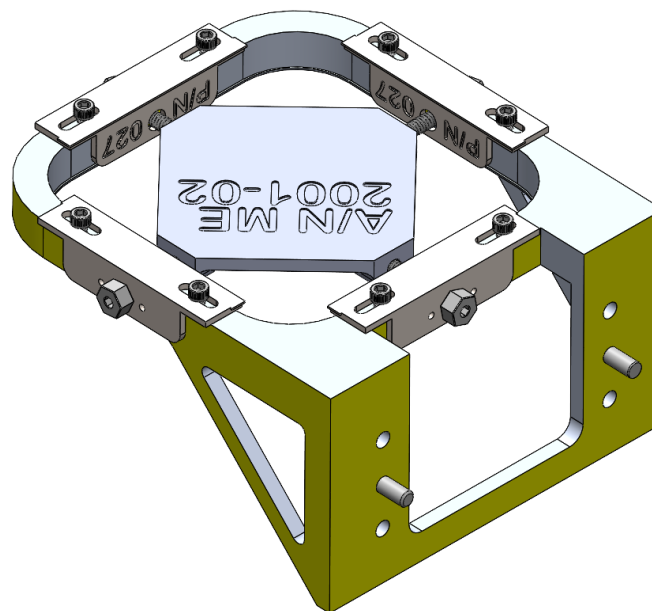


Figure 6: 3D Rendering of ASIO Shadow Plate

All current mounting interfaces between ASIO and MUSE can be found in the mechanical interface control document, ASIO-SYSE-0002. ASIO-SYSE-0002 supersedes any drawings or figures detailed in ASIO-SYSE-0001. LMSAL is to provide interfacing fasteners for flight installation as specified in ASIO-SYSE-0002, which is referenced in subsection 3.2.1.

3.1.1 Configuration Agreement Items

The items listed in Table 5 are subassembly or installation drawings that indicate interface dimensions. In the absence of an issued Configuration Agreement Item, the Solid Model configuration is the baseline. Additionally, any e-mail correspondence indicating key dimensions, in lieu of a timely drawing update, and a response of concurrence by the interfacing party is considered binding; these e-mails shall be archived in the Shared folder. The items listed in Table 5 are not required to be released to be binding. Any Configuration Agreement Item that has not been approved by the interfacing group, is by definition, not binding.

Table 5: Configuration Agreement Items.

Document Number	Document Title	Description
ME1005-001	DETECTOR ASSEMBLY	Envelope, interface dimensions, thermal hardware, field of view, and grounding interface for detector assembly
ME1005-002	SHADOW PLATE ASSEMBLY	Envelope and interface dimensions for shadow plate assembly
ASIO-SYSE-0002	MUSE - ASIO Mechanical Interface Control Document	Mechanical Interfaces, Fastener Responsibility, Surface Treatments, Mass Allocation

3.1.2 Joint Release Agreement Items

Piece part drawings indicated in Table 6 shall be submitted to the interfacing group. While formal sign-off by the interfacing group is not required, the originator of the drawing shall not circulate the drawing for local release until the item has been reviewed and acknowledgment and approval received by the interfacing party.

The information contained in these drawings shall be consistent with the relevant Configuration Agreement Items. These items need not be submitted for approval if they are entirely consistent with the Configuration Agreement Items.

The intent of this is to levy an additional interface check and to communicate tolerances that may not have been documented in interim drawings.

Table 6: Joint Release Agreement Items.

Document Number	Document Title	Description
ME1005-001	DETECTOR ASSEMBLY	Envelope, interface dimensions, thermal hardware, field of view, and grounding interface for detector assembly
ME1005-002	SHADOW PLATE ASSEMBLY	Envelope and interface dimensions for shadow plate assembly

3.1.3 As-Designed/As-Built Configuration

The documents listed below in Table 7 are unlikely to be released until hardware is built, but will be consistent with the information contained in the Configuration Agreement and Joint Release Agreement Items list. Drawings listed in Table 7 that are released prior to CDR will replace items listed in Table 5 and Table 6.

Table 7: Top Level Drawings.

Document Number	Document Title	Description
ASIO-SYSE-010	ASIO As-Built Configuration	As-Built Dimensions and Inspection of Flight Hardware

3.2 Structural Interfaces/Loads

3.2.1 Interface Fastener Responsibility

Table 8 and Table 9 directly reference document ASIO-SYSE-0002 which defines the organizational responsibility for the selection, qualification, provision, and installation of the MUSE LM/MSU interface fasteners.

Table 8: LM/MSU Interface Definitions.

Interface Number	Description
ASIO-LMSAL-01	Detector assembly to MUSE deck
ASIO-LMSAL-02	Shadow plate assembly to telescope bore
ASIO-LMSAL-03	Front plate grounding boss to MUSE grounding strap
ASIO-LMSAL-04	Back plate thermal boss to MUSE Winchester connector

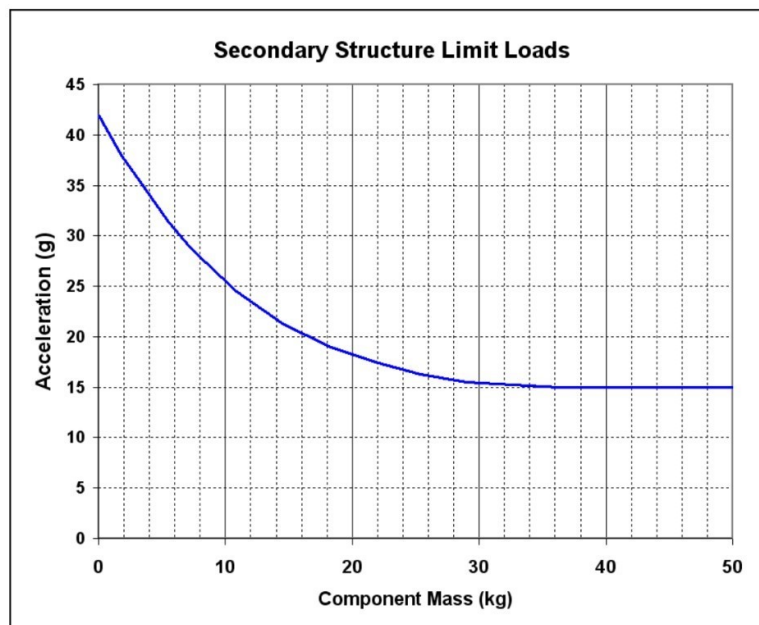
Table 9: LM/MSU Interface Fasteners Responsibility

Interface Fasteners	Defined By	Provided By	Installed By	Interface Number
6 x NAS5313E3-6, 0.1900-32 UNJF-3A, Grip Length 3/8"	MSU/ LMSAL	LMSAL	MSU/ LMSAL	ASIO-LMSAL- 01
4 X NAS1352N04-6, 4-40 UNC Socket Head Screw, Length 3/8"	MSU	MSU	MSU/ LMSAL	ASIO-LMSAL- 02
3 X NAS1351N3-10, 10-32 Socket Head, Length 5/8"	MSU	LMSAL	MSU	N/A

LMSAL is providing MSU with 3 X NAS1351N3-10 and other hardware required for ASIO-LMSAL-03/ASIO-LMSAL-004

3.2.2 Interface Loads

ASIO will be designed under the provided MUSEENV30510 for the load envelope, as seen below in Figure 7. Using an estimated lightest mass of 3.0 kg gives a design load of 35g's.



Notes: - Load is applied through the CG of the component
- Load is omni-directional

Figure 7: Limit Loads for Secondary Structure

3.3 Mass Allocation

The mass allocation is specified in the ASIO Mechanical Interface Control Document (MICD), document number ASIO-SYSE-0002.

3.4 Storage and Shipping

3.4.1 Transportation Plan & Shipping Container

ASIO will be delivered in an ESD double bag inside a Pelican (or similar) packing case. The MSU team will provide transportation protection. The MUSE team shall ensure that the ASIO instrument is maintained within a temperature range of -20°C and $+60^{\circ}\text{C}$ with a relative humidity range of 30-60% after instrument delivery and prior to launch.

The final flight filters will be shipped separately and integrated during the late stage of I&T. Exact time of integration will be decided on by MSU and LMSAL collectively. Ideally, this will be done post-vibrational testing.

3.4.2 Mechanical Handling/Storage

The ASIO instrument is to be handled with appropriate ESD precautions, with the appropriate cleanroom specified in **section 9.3**. Access to grounding points in the specified cleanroom is required for personnel during all integration activities.

ASIO is to be stored in an individual shipping container during transport, during periods between active test and integration, and while waiting to be integrated with MUSE. While in the shipping container, the DA and SPA are packaged according to the contamination control specifications and secured into the protective shipping container.

3.4.3 Replace-Before-Flight Plate

ASIO will utilize a replace-before-flight configuration during Integration and Testing. This configuration will also be utilized for aligning the Shadow Plate. The replace-before-flight configuration will be installed with captive fasteners. This configuration will be replaced with the flight configuration after testing and before launch, completing the final assembly.

3.4.4 Post-Shipment Performance Verification

The DA and SPA alignment will be verified after shipment to LMSAL. A procedure for post-shipment verification will be provided by MSU and performed by MSU personnel.

3.5 Mechanical Model Exchange

Solid Models and Finite Elements Models will be exchanged using STEP model files. MSU uses SolidWorks software.

4 THERMAL

4.1 Thermal Environments

The requirement MUSEENV31100 from document MUSE-RQ-18-0039 details the specifications for thermal design and thermal model. ASIO is designed with these specifications as governing variables.

	Cold	Nominal	Hot
Direct solar flux at 1 AU (W/m^2)	1287	1353	1419
Earth albedo (% of solar)	25%	30%	35%
Earth emission (W/m^2)	208	237	265

Figure 8: MUSE On-orbit Thermal Environment Analysis

4.2 Thermal Control Hardware

The thermal interface requirements listed below reflect the assumption that ASIO is thermally isolated from MUSE.

4.2.1 Heaters

Figure below demonstrates the location of the heater (1" x 3") with Winchester connector also visible. Location and interface specifications can be viewed in document ASIO-SYSE-0002, "MUSE - ASIO Mechanical Interface Control Document".

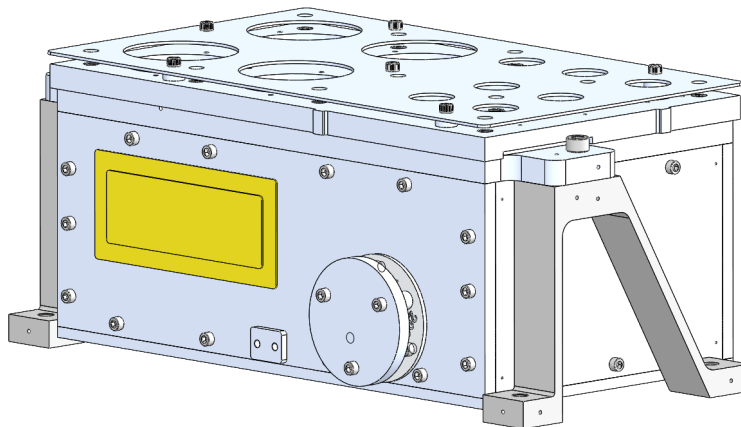


Figure 9: ASIO Heater Location

4.2.2 Insulation Blankets

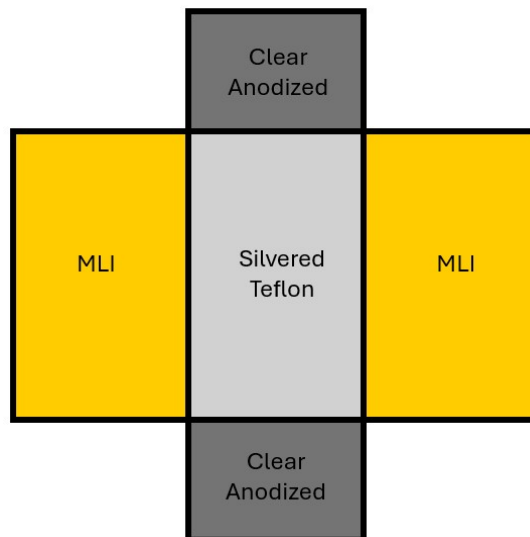


Figure 10: ASIO Surface Treatments

MLI will be necessary on the 2 long side walls of the detector assembly. MLI will be provided by LMSAL and secured with MLI specific velcro. All other surfaces will be anodized under MSU's responsibility.

4.2.3 Temperature Sensors

DA is equipped with 2 AD590 temperature sensors that are sent directly to MUSE. See section 6.1.1 for more information. Additionally DA has temperature sensors that are for MSU use only for monitoring the health of ASIO.

4.3 Responsibility for Installation of Thermal Hardware

MSU has responsibility for all internal sensors for science purposes. LMSAL is responsible for external survival heaters and a temperature sensor to go with it.

4.4 Thermal Analysis

	Part Number	Allowable Flight Limits (AFL)		Opp Model Prediction		Margin	
		Cold	Hot	Cold	Hot	Cold	Hot
HXR Diode	S3584-09	-10	50	-3.5	44	6.5	6
SXR/EUV Diode	AXUV100	-30	60	-3.5	44	26.5	16

	Part Number	Hardware Survival Limit	Opp Model Prediction	Margin
		Cold	Cold	Cold
HXR Diode	S3584-09	-20	-7.5	12.5
SXR/EUV Diode	AXUV100	-40	-7.5	32.5

Figure 11: Thermal Analysis Results and Margins (all units in deg C).

Above are the results for the thermal analysis conducted on the DA. It is predicted that ASIO can maintain within the allowable flight and hardware survival limits in hot/cold operating and a 20 minute transient cold survival lacking heat and power. Equivalent sink temperatures used in this analysis were provided by LMSAL. Allowable flight limits have been derated from the manufacturer's specification. A power dissipation of 7.5W (CBE + 40%) was assumed for the worst case hot analysis. No heater power was used for the cold case.

The following optical properties (Figure 12) were used for the analysis.

Finish	IR Emissivity	Solar Absorptivity (BOL/EOL)	Location	Reference
5mil Silvered Teflon	0.8	0.09/0.18	Sunshade	LM Thermal Handbook F5
Clear Anodized Aluminum	0.82	0.4/0.5	Short side walls	LM Thermal Handbook
MLI	0.03 effective	N/A	Long side walls	Gilmore, Spacecraft Thermal Control Handbook
Polished Aluminum	0.05	0.15	Detector filters	LM Thermal Handbook
Heavily Oxidized Aluminum	0.30	0.13	Detector filters	J. Henninger, NASA Ref. Pub. 1121

Figure 12: Planned Testing Margins Based on Thermal Analysis Results

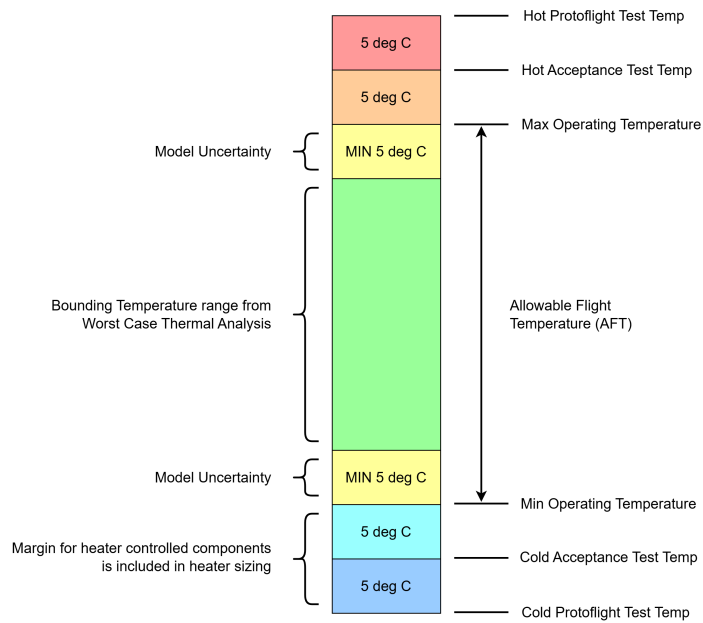


Figure 13: Planned Testing Margins Based on Thermal Analysis Results

4.5 Thermal Model Exchange

It is expected that LMSAL will provide the current best estimate of thermal conditions.

5 SOFTWARE

5.1 Data Interface

- Packet Size: 950 Bytes
- BAUD rate: 38.4 kBaud
- Communication Protocol: RS-422
 - 8 Bit words
 - No parity bit
 - 1 Stop bit
- Endianness: Little Endian

One packet will be sent to MUSE every 0.5 seconds

5.1.1 CRC Details

The CRC is a 16-bit XOR checksum (CRC16-ANSI) run on the entire packet - including the header ID "ASIO" and the footer ID "STOP". The CRC calculated bytes are located after the footer ID - see table 10 for more info.

CRC parameters:

- Polynomial: 0x8005
- Initial value: 0xFFFF

5.1.2 Packet Details

Table 10: ASIO Packet Content

Byte Size	Content	Description
4 Bytes	Start ID	ASCII text "ASIO"
1 Byte	Relay Configuration	Current relay configuration
10 Bytes	Temperature Data	Temperature info from the AD590 temperature sensors on the power board, two on the digital board, one on the hard x-ray board, and the internal temperature sensor of the microcontroller.
1 Byte	Bad Command Count	This is a counter that increases whenever a command is sent to ASIO without a header that is recognized by the system.
1 Byte	Padded Zeros	Zeros to maintain the size of packet
2 Bytes	Voltage data	1 Voltage measurement from the digital board's 3.3v+ bus
2 Bytes	Current data	1 Current measurement from the output of the DC to DC converter on the power board
2 Bytes	Packet Count	This is the packet count since startup. It will reset to 0 after passing 65536
1 Byte	Command Count	Command counter, incremented by the relay control command and the NOP command. Will reset to 0 after passing 255

6 Bytes	MUSE Time	4 Bytes of second info and 2 Bytes of sub-second info from, January 1st 1958 0:00:00 UTC, from MUSE.
2 Bytes	Padded Zeros	Zeros to maintain the size of packet
4 Bytes	ASIO Time	ASIO time from startup in milliseconds
4 Bytes	First Data Point ASIO Time	The time that the first set of data points was collected in milliseconds
4 Bytes	Padded Zeros	Zeros to maintain the size of packet
...		
-	Data	Repeated data segments detailed in 5.1.3
...		
4 Bytes	End ID	ASCII text "STOP"
2 Bytes	CRC	A CRC ran on the entirety of the packet. Excluding the last two bytes left for the CRC to be placed in the packet.

Blue for Header, Orange for Data, Green for Footer

5.1.3 ADC Data field of a packet

The data field of the packet will be filled with the ADC results from our channels. The packet will be populated with values in this order, SXR1, SXR2, SXR3, SXR4, HXR, EUV. Each data point is 3 bytes, totaling 18 bytes for a 10 millisecond period. This is detailed below.

Table 11: ASIO Data Segments

...		
18 Bytes	ADC Data	6 channels of data each 3 bytes (SXR1, SXR2, SXR3, SXR4, HXR, EUV)
...		

5.1.4 Commanding and Receiving Time

ASIO will have 3 commands each prefaced by a hexadecimal code, some codes will expect a parameter. More details on the commands are given in Table 12. Additionally, this table details what data needs to be sent in order to be recognized as a command. Any commands sent without one of these headers will pass through the code and increase the bad command count. Only the Relay and No Operation commands will increment the command count.

Table 12: Command Structures

Commands	Hex Code	Following Parameters	Description
Incoming time	0xAD	6 bytes of MUSE time.	Incoming time-at-tone packet corresponds to the pulse per second. See section 5.1.6. Big endian, containing four bytes of seconds data and two bytes of sub-second data since the epoch.
Relay Control	0xCB	1 Byte of Relay configuration then 5 bytes of padded zeros	Change the configuration of the relays. See the example below.
No Operation	0xCA	6 bytes of padded zeros	NOOP command, only increments the received command counter

All commands are 7 bytes

Example Time Command,

Hex Value	0xAD	0x69	0x0B	0xA0	0x54	0x16	0x73
Order Recieved by ASIO	1	2	3	4	5	6	7
Significance	-	MSByte	-	-	LSByte	MSByte	LSByte
Data Type	Header	Seconds	Seconds	Seconds	Seconds	Sub-Seconds	Sub-Seconds

The least significant bit will be transmitted first for each byte.

Example Relay Control Command (1 byte of relay configuration)

EUV5	EUV6	EUV7	HXR8	HXR9	HXR10	HXR11	Unused
1	1	1	0	1	0	1	0
LSBit	-	-	-	-	-	-	MSBit

HXR8 and HXR10 to be turned off. The least significant bit, to be received by ASIO first, is the bit representing EUV5. Relay numbers correspond to the photodiode orientation found in Figure: 4

For each relay command that is sent, all relays are turned on. Once all relays are on, only the ones specified in the command will be turned off.

5.1.5 Epoch

MUSE time commands contain the time since January 1, 1958, 00:00:00 UTC.

5.1.6 Pulse Per Second

Time data will be sent to ASIO once per second, as shown in Figure 14. The PPS line must be set to a logic LOW at least 125 ns before the next rising edge, and the PPS line must be set to a logic HIGH for at least 125 ns before it can return to the LOW state.

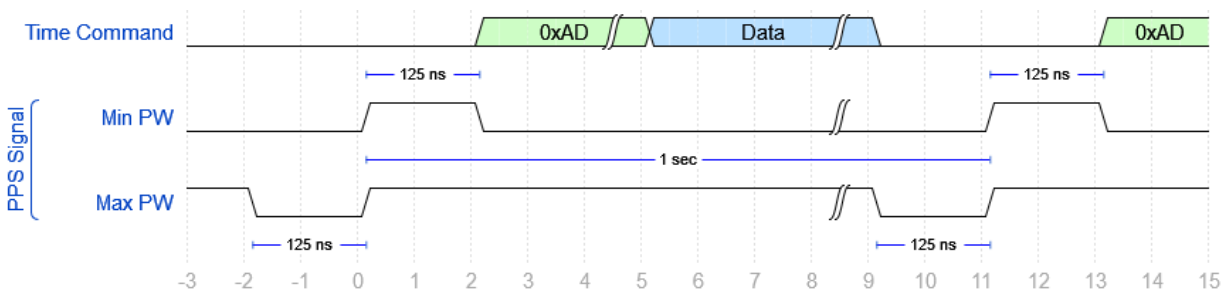


Figure 14: PPS Pulse Width Timing Diagram

A time command should be sent after every PPS Pulse. This time command will contain the 6 bytes of MUSE time data corresponding to the rising edge of the previous PPS Pulse. In Figure 14 above, the blue 'Data' field should contain the time information of the rising edge at position 0.

This time command can be sent by MUSE any time after the corresponding PPS rising edge, but the transmission must be complete before the next PPS rising edge.

If ASIO does not receive a time packet after a given packet, ASIO will carry on with no errors.

Note: The PPS line will actually be a differential signal. This timing diagram displays logic levels of the signal, not the voltage on a line.

5.2 Testing Plan

Refer to Document ASIO-TEST-0027 for the detailed testing plan.

5.2.1 Conversion Formulas

All of the values outputted in the packet are the raw values received from the various peripherals. In order to get the real values for some of the data a conversion will need to be performed on the raw value. The calculation is outlined in the table below.

Table 13: Conversion Formulas

Data	Size (bits)	Conversion Formula	Units
Power AD590 Temperature	16	$[K] = integer * \frac{3.3}{2^{12}} * 1000$	Kelvin
Digital 1 AD590 Temperature	16	$[K] = integer * \frac{3.3}{2^{12}} * 1000$	Kelvin
Digital 2 AD590 Temperature	16	$[K] = integer * \frac{3.3}{2^{12}} * 1000$	Kelvin
HXR AD590 Temperature	16	$[K] = integer * \frac{3.3}{2^{12}} * 1000$	Kelvin
MCU Internal Temperture	16	$[K] = \frac{(\frac{3}{3.3} * integer) - 669}{2.275} + 303.15$	Kelvin
Digital Bus Voltage	16	$[V] = \frac{integer}{800}$	Volts
Digital Current Draw	16	$[A] = \frac{integer}{250}$	Amps
SXR Channels	24	$[A] = \frac{(integer * \frac{2}{2^{23}}) + 2.5}{180E6}$	Amps
HXR Channel	24	$[A] = \frac{(integer * \frac{2}{2^{23}}) + 2.5}{50E6}$	Amps
EUV Channel	24	$[A] = \frac{(integer * \frac{2}{2^{23}}) + 2.5}{56E6}$	Amps

All values sent in Little Endian Format

6 ELECTRICAL

6.1 Electrical Interfaces

ASIO will have three external connectors, organizing the signals into three categories: Power, Communication, and Programming. Each subsection below corresponds to each of the three connectors and provides details about the component and signal pin location.

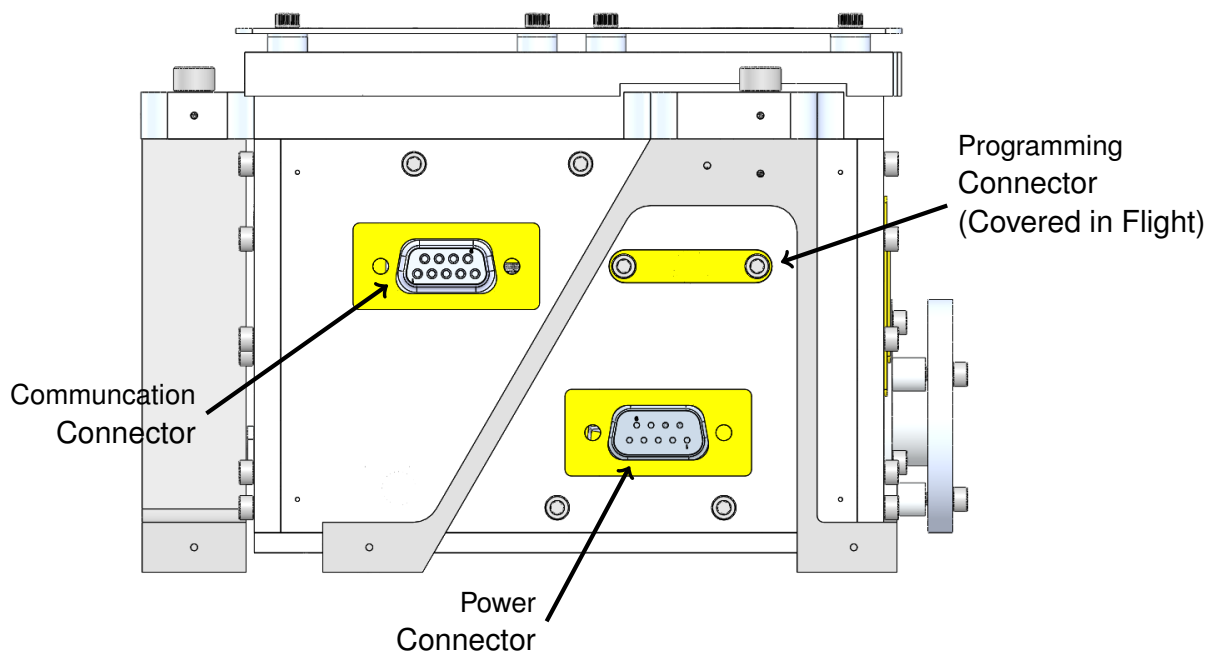


Figure 15: ASIO Connector Interface

6.1.1 Power Connector (J1)

Component Specifications		
Item	ASIO Component	MUSE Component
Part Number	311P409-1P-B-12	311P409-1S-B-12
Pin/Socket Part Number	G10P1	G10S1
Number of Pins	9	9
Grade	Level 1	Level 1
Shell Finish	Gold	Gold

Table 14: Power Connector Component Specifications

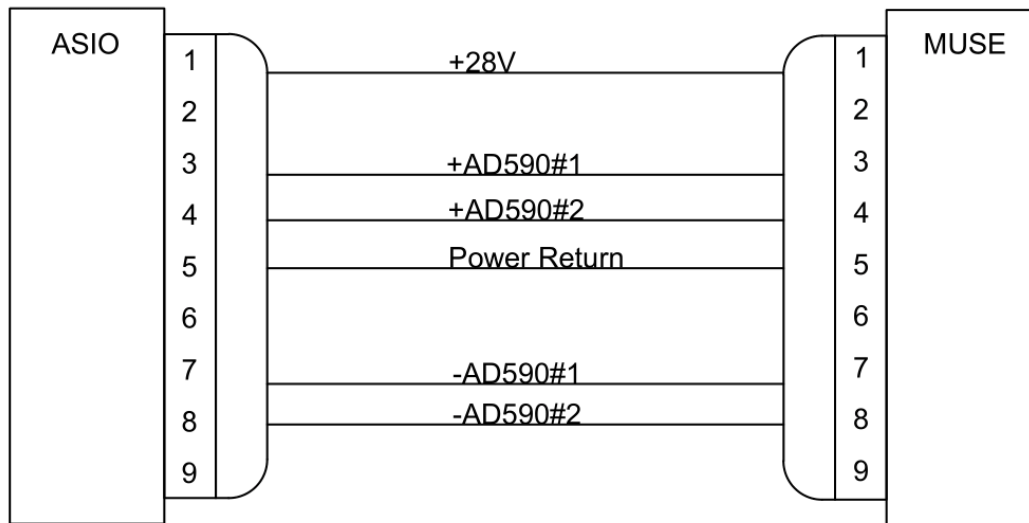


Figure 16: Power Connector Interface Diagram

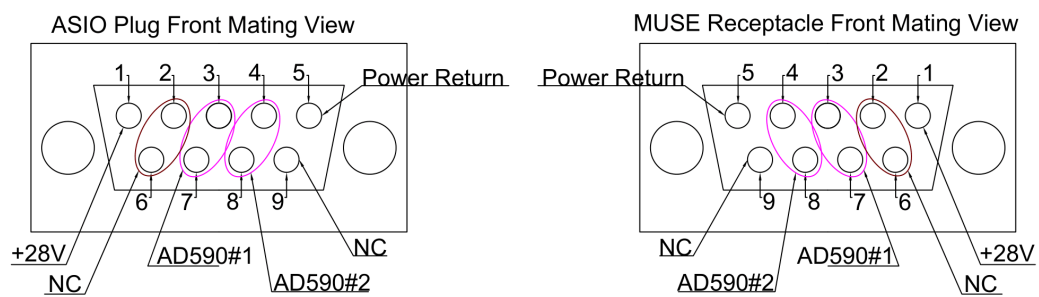


Figure 17: Power Connector Pin Diagrams

Signal Summary		
Signal Name	Description	Pins
+28V	Main +28V Power Bus	1
Power Return	Main Power Return Bus	5
+AD590#1	Positive AD590 Terminal	3
-AD590#1	Negative A D590 Terminal	7
+AD590#2	Positive AD590 Terminal	4
-AD590#2	Negative AD590 Terminal	8
NC	NO Connection	2, 6, 9

Table 15: Power Connector Signal Summary

The AD590 temperature sensors IC will be located inside ASIO, one will be on the DC/DC converters, and the other will be on one of the inside walls. The positive and negative leads will be directly connected to MUSE, and will NOT be electrically connected to ASIO anywhere.

LMSAL will build the required cable to properly connect the ASIO Power Connector to MUSE. This cable will have **AWG 24** size wires.

MUSE will implement a slow start circuit on the main +28V power bus.

See document number MUSE0001449 for a detailed circuit schematic of the ASIO power system.

6.1.2 Communication Connector (J2)

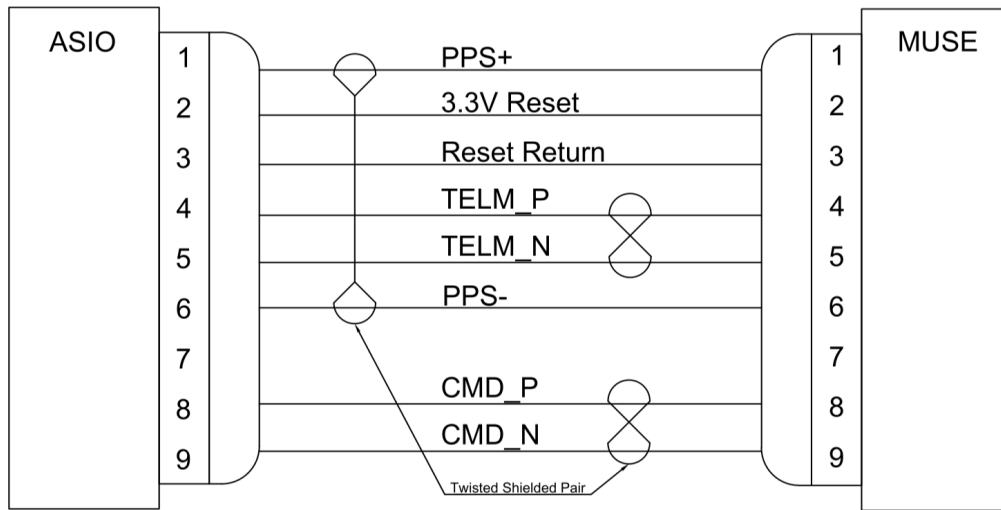


Figure 18: Communication Connector Interface Diagram

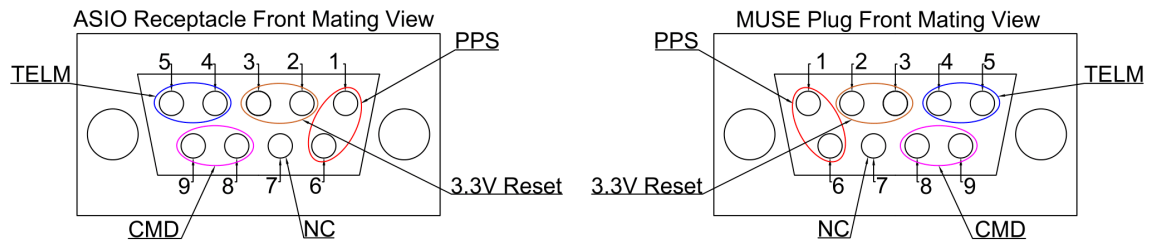


Figure 19: Communication Connector Pin Diagrams

Component Specifications		
Item	ASIO Component	MUSE Component
Part Number	311P409-1S-B-12	311P409-1P-B-12
Number of Pins	9	9
Pin/Socket Part Number	G10S1	G10P1
Grade	Level 1	Level 1
Shell Finish	Gold	Gold

Table 16: Communication Connector Component Specifications

Signal Summary		
Signal Name	Description	Pins
TELM_P	Positive Telemetry Signal	4
TELM_N	Negative Telemetry Signal	5
CMD_P	Positive Command Signal	8
CMD_N	Negative Command Signal	9
PPS+	Positive Pulse Per Second Signal	1
PPS-	Negative Pulse Per Second Signal	6
Reset	Positive 3.3V Reset Signal	2
Return	Reset Signal Return	3
NC	NO Connection	7

Table 17: Communication Connector Signal Summary

Detailed Signal Description	
TELM	The differential Telemetry lines (TELM) will be used to transmit instrument data from ASIO to MUSE. A full data packet will be sent each time the packet is assembled by the ASIO system. See Figure 20 for the termination diagram.
CMD	The differential Command lines (CMD) will be used to send information from MUSE to ASIO. A command can be sent by MUSE to ASIO at any time. See Figure 20 for the termination diagram.
PPS	The differential PPS (Pulse Per Second) signal line will be used to align a timestamp sent via the Command lines with the true time. The PPS line will be set to a logic 1 once per second to align the corresponding time information. ASIO will treat the rising edge of the pulse as the true time of the received timestamp information. See Figure 14 for more info.
Reset/Return	The reset and return signals will be used to soft reset ASIO in the event of a system failure. See section 6.3 for more details.

Table 18: Detailed Signal Description

For more information about the commanding and time alignment see **section 5.1.4**.

The RS-422 CMD and TELM lines will need a termination load on the receivers. The following Figure describes the termination circuit for proper communication with ASIO.

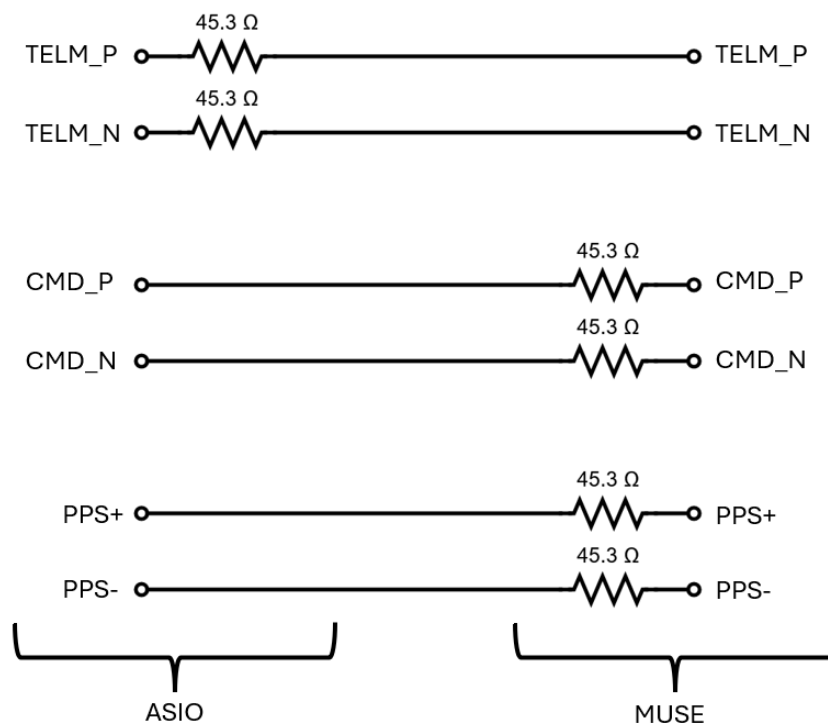


Figure 20: RS422 Termination Circuit Diagram

The shielding on all twisted and shielded pairs will be electrically connected to the ASIO chassis.

LMSAL will make the cable required to connect between ASIO and MUSE. This cable will have a twisted shielded pair for each differential signal (TELM, CMD, and PPS) as detailed above.

6.1.3 Programming Connector (J3)

ASIO can be programmed with both the JTAG and Serial Wire protocol. This connector will be covered in flight with an aluminum plate and will **NOT be connected to MUSE**. This plug will be used by the MSU team for programming, testing, and debugging only. A pin diagram for this connector is shown below.

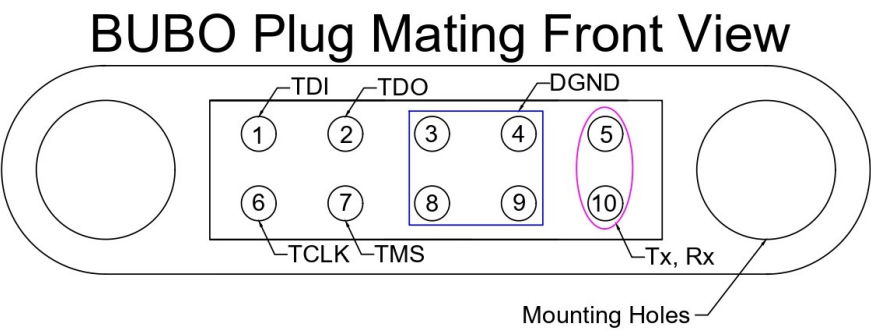


Figure 21: Programming Connector Pin Diagram

Component Specifications	
Item	ASIO Component
Part Number	G125-MC11005M1-0150L
Number of Pins	5+5
Manufactured By	Harwin

Table 19: Programming Connector Component Specifications

Signal Summary		
Signal Name	Description	Pins
TDI	Test Data In	1
TDO	Test Data Out	2
TCLK	Test Clock	6
TMS	Test Mode Select	7
Tx	Data Transmit	5
Rx	Data Receive	10
DGND	ASIO Digital Ground	3, 4, 8, 9

Table 20: Programming Connector Signal Summary

6.2 Grounding

ASIO will have two designated electrical grounds. The Digital Ground (DGND) will be connected to the chassis. The Analog Ground (AGND), separated to improve signal integrity, will be internally isolated and NOT connected to MUSE through a connector pin. The Analog Ground will be connected to the Digital Ground at one point inside ASIO.

LMSAL will make a copper Grounding Strap to ground the chassis of ASIO to the MUSE spacecraft deck. See **Section 3.1** for the Grounding Strap mechanical interface.

Power Return signals will be isolated from all other grounds.

6.3 Soft Reset

The ASIO microcontroller software can be reset from the 3.3V reset lines shown in Figure 19. The pulse width specifications are shown below in Figure 22.

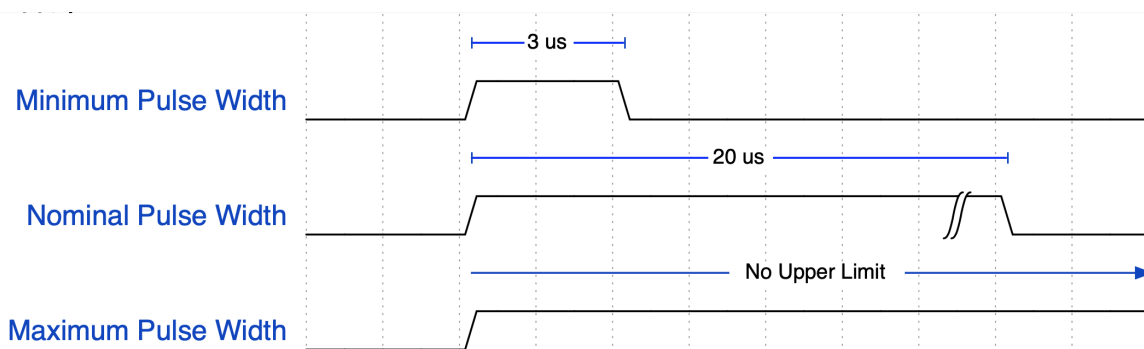


Figure 22: Soft Reset Pulse Width Diagram

The reset pulse width should be near 20 us, the nominal pulse width, but can be longer if needed. The pulse width should not be made to be shorter than 3 us. The reset lines must stay at a logic LOW when ASIO is in normal operation.

The circuit diagram for the ASIO reset circuit is shown below in Figure 23. This circuit uses an optocoupler to isolate the MUSE electronics from the ASIO electrical system.

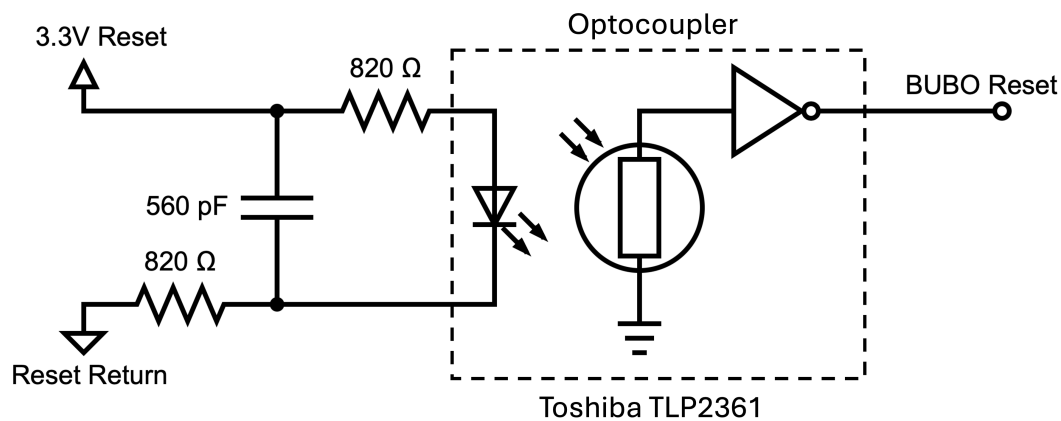


Figure 23: Soft Reset Circuit Diagram

6.4 Comprehensive Performance Test

ASIO will perform a comprehensive performance test (CPT), as defined in ASIO-SYSE-0014, during various stages of integration and testing. Data from this CPT will be analyzed by ASIO team members to determine instrument status.

7 Integration Management

Includes all interfacing processes and requirements for ASIO to integrate with MUSE successfully. All conditions and processes are to be agreed upon by MSU and LMSAL.

7.1 Areas That Require Access for Integration and Testing

Several areas are required for access for integration and testing after delivery to LMSAL.

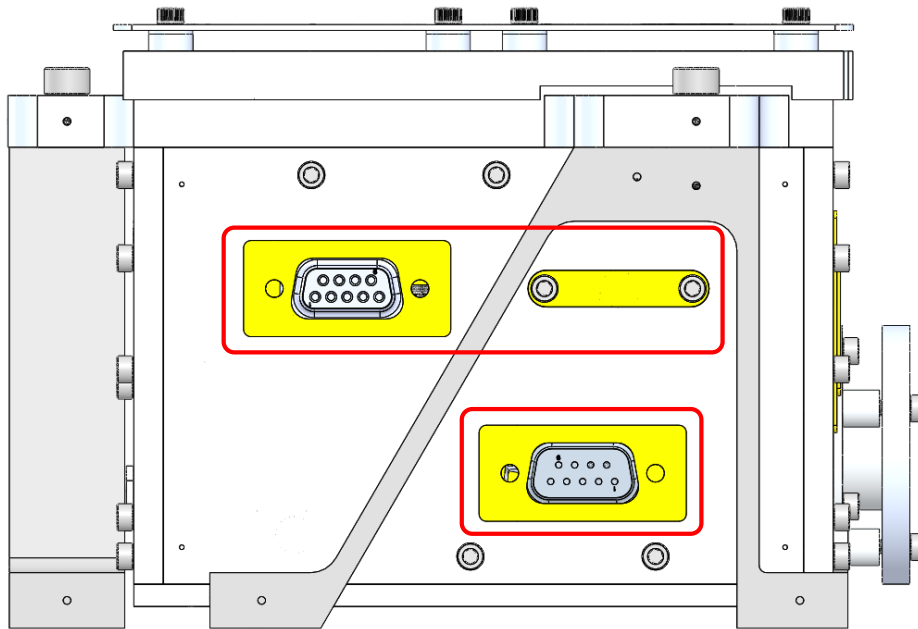


Figure 24: Area Required on the Side of Detector Assembly

Figure 24 is required to be accessible for implementation with MUSE. Integrating the connectors is required for implementation with MUSE. More information about the connectors is detailed in section 6.

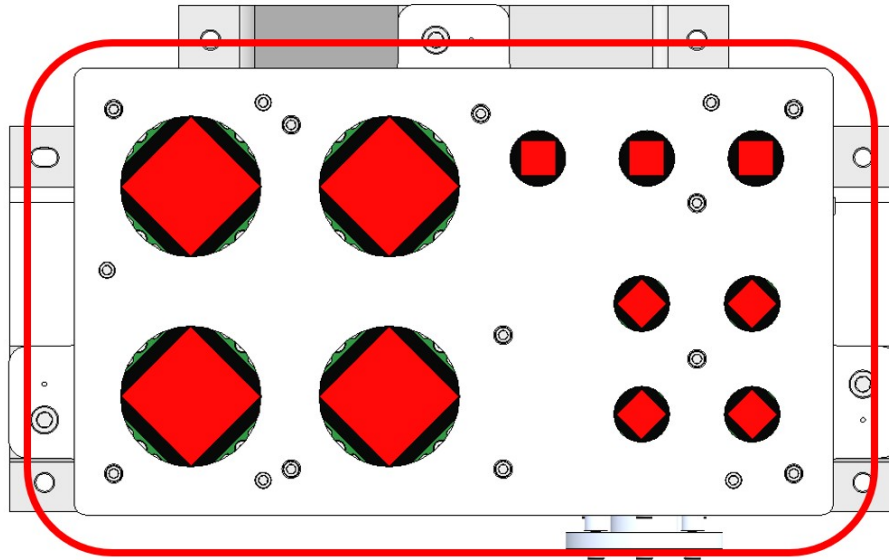


Figure 25: Area Required on the Top of Detector Assembly

The top of the Detector Assembly is required to be accessible in order to change the configuration as detailed in subsection 7.2.

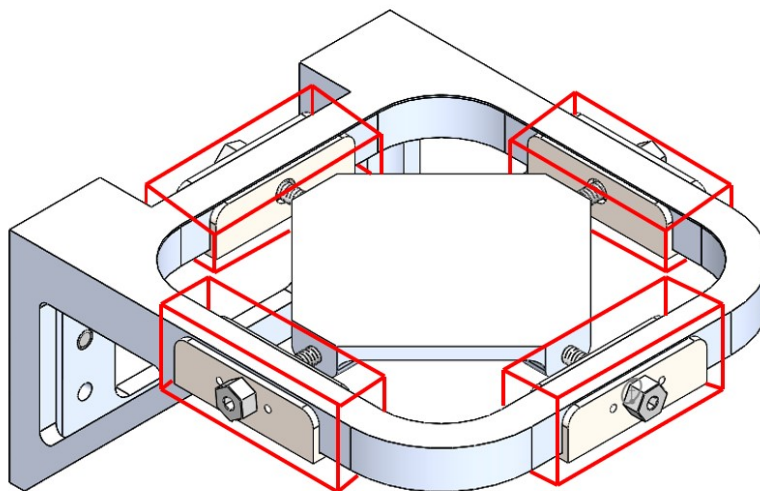


Figure 26: Area Required on the Shadow Plate Assembly

Figure 26 illustrates the accessibility required. The Shadow Plate will be adjusted to achieve alignment with the detector assembly. Access to the face of the fasteners and the steel washers are required for adjustment and liquid pinning for the final alignment. The alignment of the

Shadow Plate Assembly is detailed in subsection 7.4

7.2 Configuration Management

Table 21: Configuration Management Table

Configuration	Description	Purpose
Replace Before Flight	Surrogate filter plate (Red anodized aluminum), handles, and surrogate sun shade	Used for delivery, alignment, vibe and thermal vacuum testing
Flight	Filter plate with filters and sun shade installed	Installed as late as possible and only used for flight

The configurations described in Table 21 are displayed in Figures 30 and 28, respectively. Further configuration guidance is provided in subsection 7.3.

7.3 Configuration Requirements and Assembly

Both configurations are composed of the specified configuration plate, sun shade, and the specified fasteners. The following documentation supports the specified requirements and assembly of each configuration.

Each configuration will be pre-assembled with the exception of the fasteners required to interchange the configurations. Figure 27 demonstrates the fasteners required to change out each configuration. Note that this image uses the Flight Configuration. The sun shade is equipped with pass through holes for the fasteners to secure each configuration. The table below details the fasteners required for configuration change.

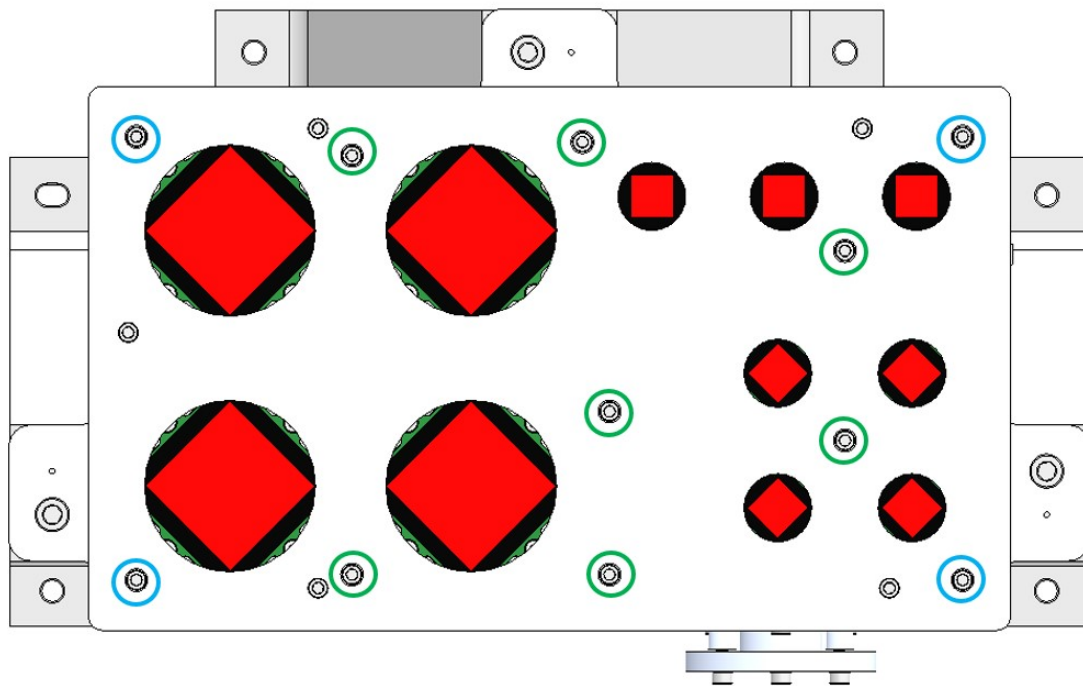


Figure 27: Fastener Locations for Changing Configurations

Table 22: Fasteners for Configuration Requirements

Color	Fastener Description	Specifications	Quantity
Blue	High Strength A286 SS Socket Head, 4-40 Thread Size, 3/4" Long	Modified NAS1352N04-12	4
Green	High Strength A286 SS Socket Head, 4-40 Thread Size, 3/8" Long	Modified NAS1352N04-6	7

The fasteners listed in Table 22 will be supplied by MSU and are shown for reference only.

7.3.1 Flight Plate

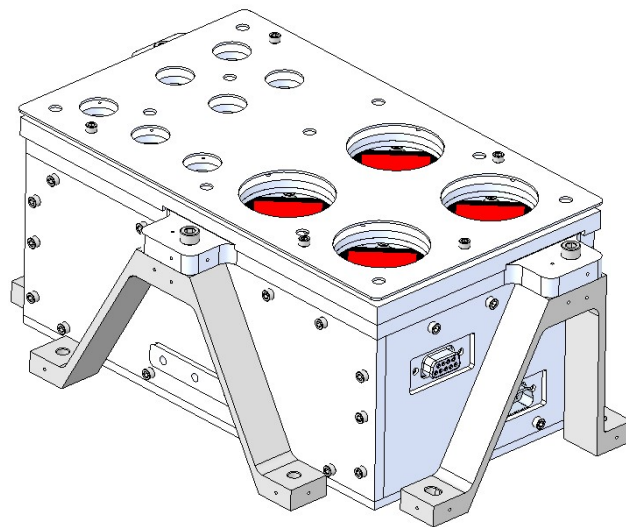
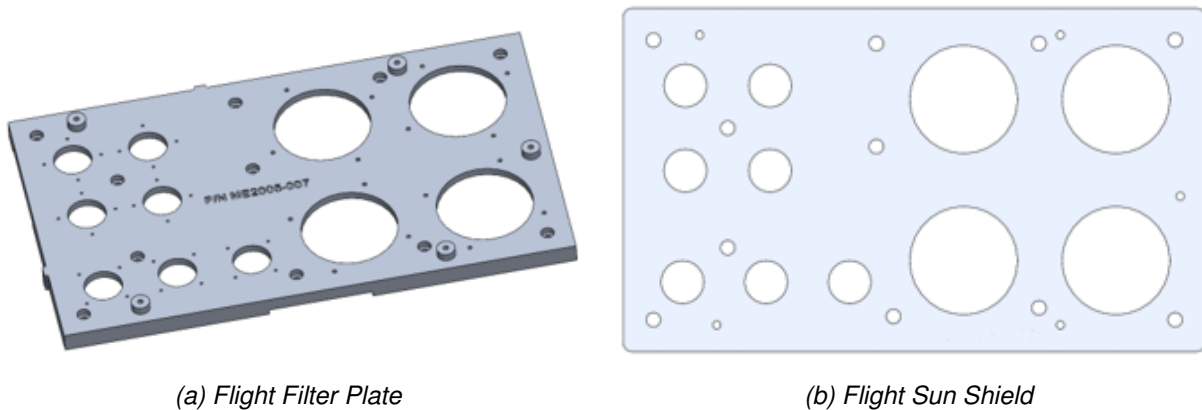


Figure 28: Detector Assembly Flight Plate Configuration

Figure 29 illustrates the filter plate and sun shield used for the Flight Plate assembly.



(a) Flight Filter Plate

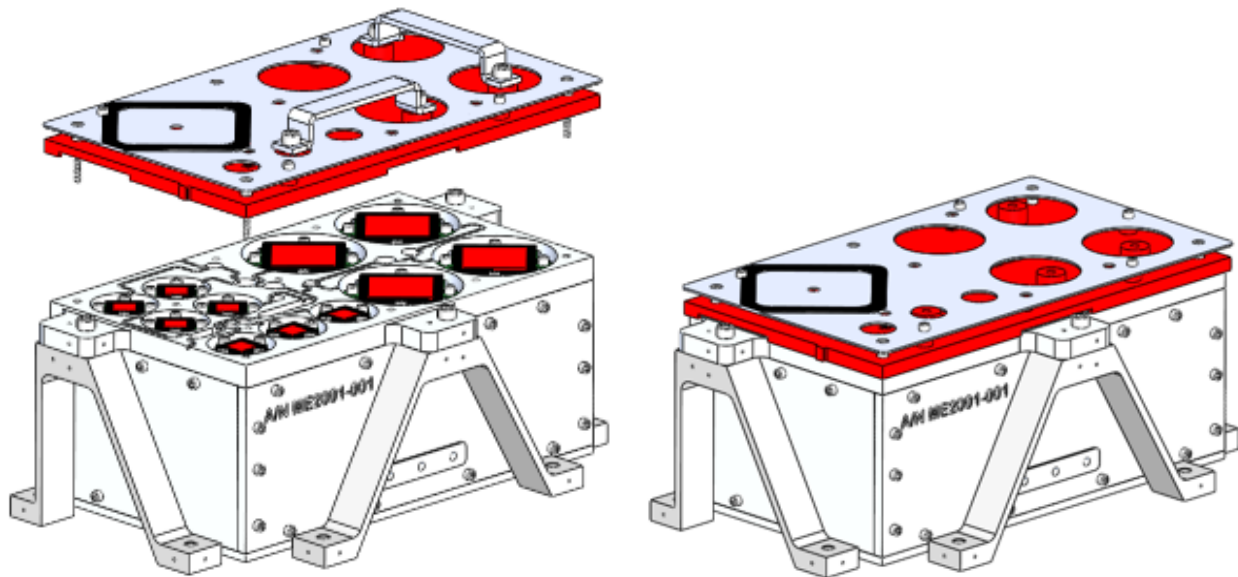
(b) Flight Sun Shield

Figure 29: Flight Configuration Parts

7.3.2 Replace Before Flight Plate

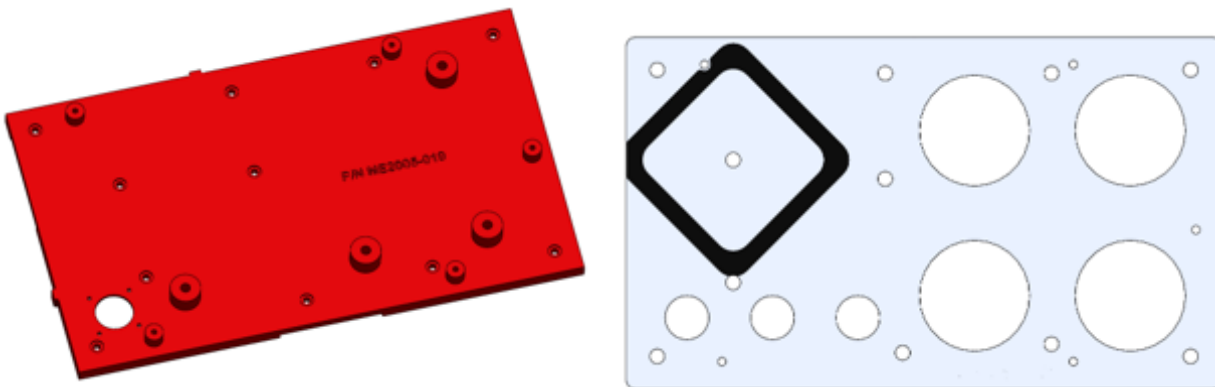
Figure 30 illustrates the configuration for the Replace Before Flight Plate assembly. The fasteners in the replace before flight plate will be captive fasteners. This includes the fasteners in the handles, since the handles will be removed for testing. The configuration with handles (Figure 31a, displayed in exploded view) will be used for transportation and when replacing

with the flight plates. The configuration without handles (Figure 31b) will be used for all alignment and testing. Figure 31 displays the versions of the Filter Plate and Sun Shield that will be utilized for the Replace Before Flight configuration.



(a) Replace Before Flight Configuration with Handles (b) Replace Before Flight Configuration without Handles

Figure 30: Replace Before Flight Configuration



(a) Replace Before Flight Filter Plate

(b) Replace Before Flight Sun Shield

Figure 31: Replace Before Flight Parts

7.4 Alignment of Shadow Plate

The SP alignment requirements and procedure are specified in the Shadow Plate Alignment Requirements and Procedure, document number ASIO-SYSE-0011.

8 DESIGN FOR DEMISE

The design of the ASIO Assembly shall consider the design for demise guidelines, A101-RP-10-0765. Most notable is to avoid the use of, and/or amount of titanium where practical. The ASIO Assembly shall contain zero parts composed of titanium, stainless steel, glass, or ceramics larger than 1 kg mass each.

9 CONTAMINATION CONTROL

Contamination control of ASIO Hardware at LMSAL is specified in MUSE-PN-18-0078.

9.1 Material Outgassing

MSU will use low outgassing materials. Low outgassing is defined as less than 1% TML, and less than 0.1% CVCM as defined in MUSE-PN-18-0066. If non-outgassing materials are needed, they will need to be approved by LMSAL and the ASIO principal investigator. Thermal bakeouts, verification, and analysis shall be performed to ensure that materials used in ASIO's hardware have been conditioned to reduce the possibility of introducing harmful levels of molecular contamination.

ASIO will undergo a vacuum thermal bakeout to prevent the buildup of molecular contamination. The thermal bakeout will be governed by a time at temperature in the absence of MSU owning a TQCM. The bakeout will occur on the flight unit at 60°C for at least 96 hours. MSU will provide LMSAL with a full expected and as-built materials list.

9.2 Venting Requirements

ASIO is designed to vent any internal volumes away from sensitive MUSE components. MSU performed a venting analysis to verify pressure differentials and venting locations required from ASIO to minimize cross-contamination. ASIO vent locations are defined in Figure 32.

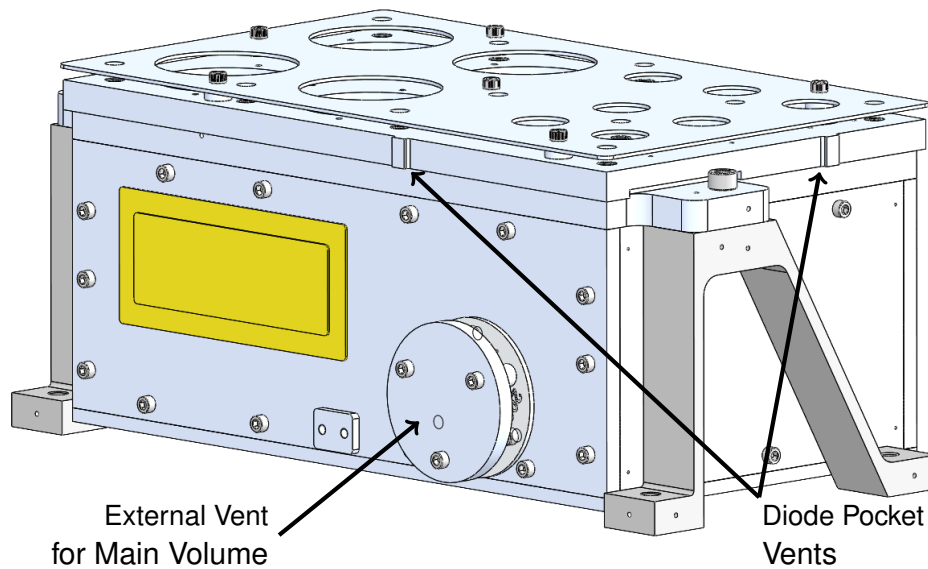


Figure 32: ASIO Vent Locations

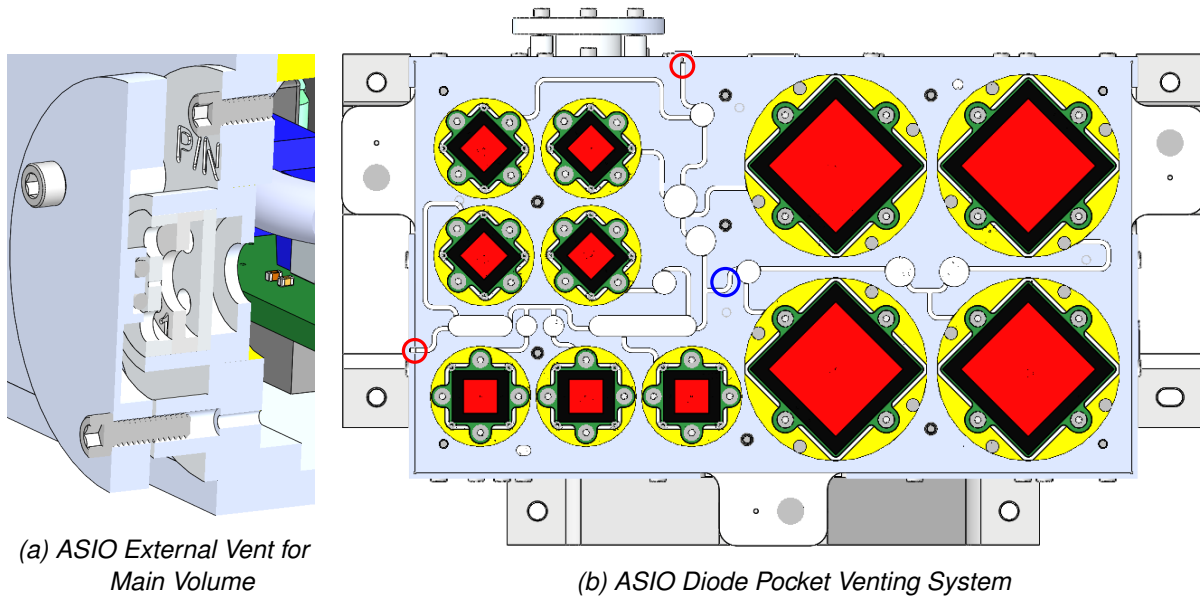


Figure 33: ASIO Venting Systems

The Luxel filters are sensitive to exposure to pressure differentials. The filters can not experience a pressure differential greater than 1 torr (0.02 psid). MUSE-PN-18-0078 specifies that the Instrument and Spacecraft will be designed to maintain the maximum delta-pressure below 0.1 psid and allow proper molecular outgassing of enclosed volumes on orbit. Additionally, MUSE-RQ-18-0039 specifies that the MUSE observatory shall be compatible with the depressurization rate of 0.7 psi/sec.

The design of the venting system equipped on ASIO meets the specifications above. Table 23 details the total volume divided by the area of the vent for the External Vent and Diode Pocket Venting System. The internal volume of the detector assembly is 100.48 cubic inches. Further analysis is displayed in ASIO-SYSE-0015.

Table 23: External Vent Analysis

Part	Description	V/A (in)
Vent housing built-in baffle	Diameter of 0.26 in, 1 hole	1892.68
Baffle #1	Diameter of 0.13 in, 4 holes	1892.68
Baffle #2	Diameter of 0.26 in, 1 hole	1892.68
Baffle #3	Diameter of 0.13 in, 4 holes	1892.68

Table 24 describes the venting analysis completed for the diode pockets. The first row displays the calculations for the area indicated by a blue circle in Figure 33b. This calculation is to ensure adequate venting for the three HXR photodiode pockets that will vent through that area. The second row includes the calculations for the areas indicated by red circles in Figure 33b. This calculation accounts for all seven photodiode pockets, as well as having two vents. Further analysis is available in ASIO-SYSE-0015.

Table 24: Diode Vent Analysis

Part	Description	V/A (in)
3xHXR	Radius of 1.063 in	447.96
4xHXR; 7xSXR/EUV	4xRadius of 1.063 in 7xRadius of 0.586 in	553.73

Using the information indicated by Figure 34 it can be determined that the venting system is sufficient to depressurize at 0.7 psi/sec and the maximum pressure differential allowed for the Luxel filters.

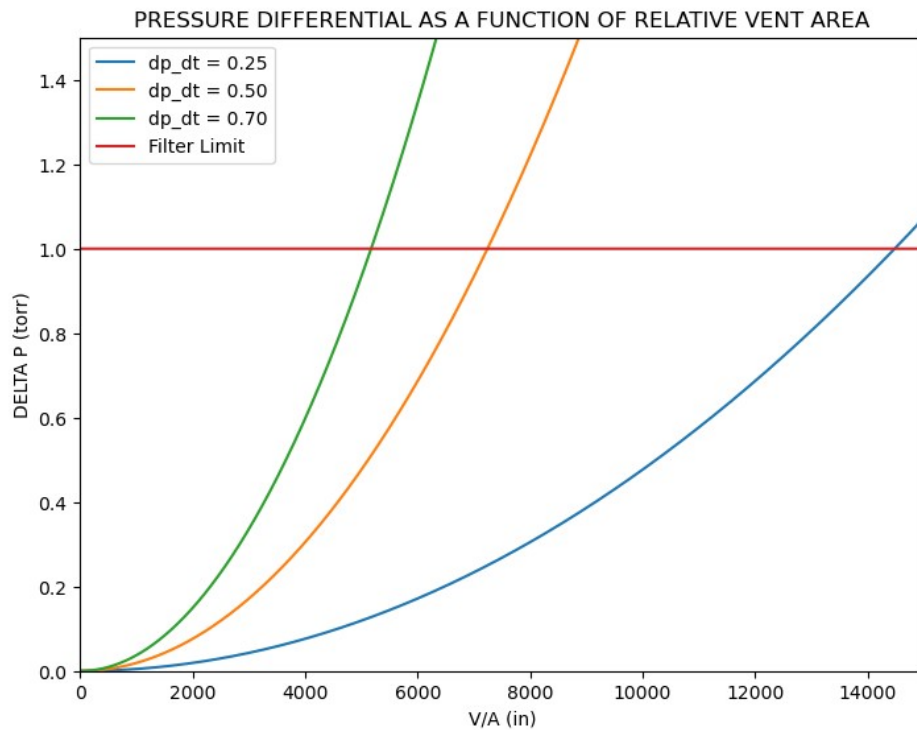


Figure 34: ASIO Venting Analysis

The venting strategy for ASIO will be reviewed and approved by the MUSE Project and MSU so that all requirements are met.

9.3 Processing Environment

ASIO requires a Class 8 (Class 100,000) or lower cleanroom for the processing of the basic assembly. Critical hardware such as the processing and installation of filters require a Class 7 (Class 10,000) or lower cleanroom. Cleanrooms are required to adhere to the guidelines outlined in MUSE-PN-18-0078. To prevent any damage to ASIO's sensitive electronics precautions beyond contamination control will be required. The use of antistatic packaging, ESD-approved garments, and grounded wrist straps.

9.4 Packaging and Transportation Material Requirements

ASIO will adhere to MUSE-PN-18-0078 packaging material requirements by using approved bagging. Additionally, Packaging will include a clean electrically conductive laminate as the first barrier and Nylon as the second barrier to maintain the same level of cleanliness. The outer wrap will be composed of polyethylene or another approved material outlined in MUSE-

PN-18-0078. During packaging and transportation, there will be clear identification of contents and cleanliness levels. For transportation, ASIO will be transported following the packing technique outlined in MUSE-PN-18-0078 and with the use of a protective shipping container.

10 VERIFICATION

Hardware produced by MSU shall be verified in accordance with the MUSE Environmental Test Matrix (MUSE-RQ-23-0342). The DA and SPA hardware will be shown to conform to the Configuration and Joint Agreement item specified herein and to MSU Interface Drawing MUSE-400 (TBD).